This Week

1. [Literature] Paper presentation: ReNorm
2. [Literature] Paper presentation: CLIP-FDGI
3. [Sweep] Omega Regularization Sweep
4. [Model] Attempts at fixing the SLERP / analytical distance computation
5. [Test] Benchmarked the Finsler BAU against other papers in the same field
6. [Model] Domain-Conditioned Global Drift

Tabled for next week

1. [Model] Devise a plan to migrate Finsler modifications to CLIP-FDGI / ReNorm
2. [Visualisation] Plot the omega drift vector
3. [Model] Manifold Smoothness Regularization: Force the network's vector field to match the surrogate.
   - During training, sample the midpoint of the geodesic $\gamma(0.5)$
   - Pass this midpoint through the neural network to obtain the true intermediate drift vector ω_{true}
   - Compute the parallel-transported surrogate vector at the midpoint, $\omega_{surrogate}$.
   - Add a regularization loss: $L_{smooth} = \| \omega_{true} - \omega_{surrogate} \|_2^2$

[Literature] ReNorm

ReNorm: Re-evaluating Instance Normalization for DG Re-ID

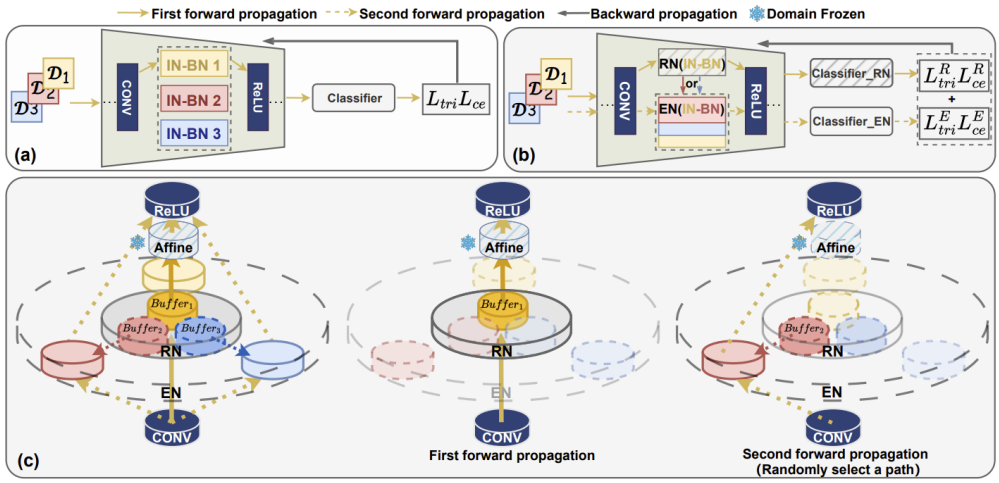
- **The Problem:** Standard Instance Normalization (IN) indiscriminately flattens feature variances, inadvertently destroying fine-grained identity structures (e.g., clothing textures) alongside domain styles.
- **The Solution:** Introduces an adaptive, channel-wise regularization mechanism to modulate variance reduction.
- **Mechanism:** Identifies and mathematically protects variances highly correlated with identity discrimination, while standardizing only those strictly tied to domain style.

Comparing with BAU:

- **Strategy:** ReNorm operates at the *architectural feature-map* level (manipulating mean/variance statistics prior to pooling), whereas BAU operates entirely at the *output embedding metric* level.
- **Philosophy:** ReNorm believes domain gaps are best handled by selectively removing style at the CNN/Transformer intermediate layers. BAU believes domain gaps are best handled by repelling domain clusters directly on the unit hypersphere.

- *Context:* In typical Domain Generalization for Re-ID, networks try to filter out domain-specific noise.
    - Methods like IBN-Net use standard normal distributions $N(0, 1)$ to achieve this.
- *Elaboration:* ReNorm shows that flattening σ^2 is destructive.
    - It proposes an attention-like parameter λ_c to selectively scale the normalized feature, retaining the residual identity signal.
- *Connection to our work (BAU/Finsler):*
    - ReNorm highlights a critical flaw in traditional DG Re-ID: treating domain bias purely as noise to be erased harms retrieval.
    - Our Finsler approach takes the opposite, constructive route
        - we explicitly isolate the domain bias into an orthogonal drift slice x_{drift} rather than attempting a destructive normal mapping.

[Literature] ReNorm



Directing Attention (The Standard Approach):
top left shows how traditional domain-generalizable Re-ID networks handle feature normalization.

- They typically interleave Batch Normalization (BN) and Instance Normalization (IN) layers.
- As the text snippet confirms, while BN helps with discriminative features, it overfits to source domains. IN is used to eliminate that domain-specific style.

The Critical Flaw:
The core problem that ReNorm highlights is:

- treating domain bias purely as noise to be erased harms retrieval.
- When standard IN standardizes the feature variances to remove 'domain style',
- it wipes out the fine-grained, discriminative identity structures—like clothing textures—that are crucial for distinguishing between different people.

The ReNorm Solution (The Dual Path):
To solve this, ReNorm steps away from static IN-BN blocks.

- As shown in the right and bottom graphics, it uses a unique dual forward-

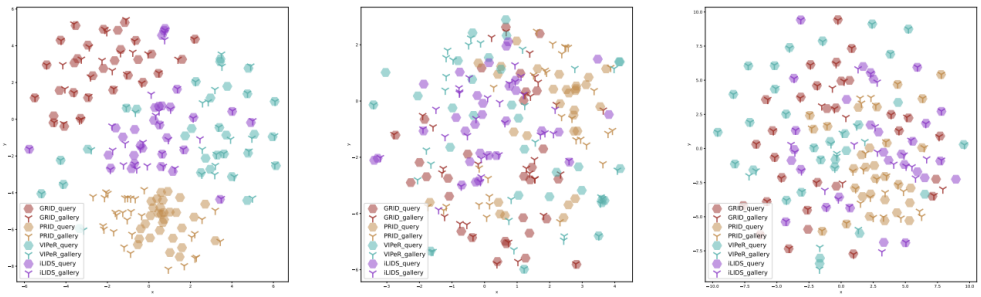
propagation strategy with the exact same network weights.

- In the first pass, it uses **RN (Representation Normalization)** to focus purely on extracting the most highly discriminative identity features within the source domains.
- In the second pass, it uses **EN (Exploration Normalization)** to actively force those weights to generalize to unseen target domains."

Our Narrative Takeaway:
ReNorm illustrates that destroying domain information at the feature level is mathematically highly destructive to identity tracking.

- While ReNorm attempts to fix this by balancing two complex normalization paths, our repository takes a cleaner geometric approach:
- we don't try to destroy the domain bias at all. Instead, we explicitly isolate it into a dedicated Finsler drift vector, preserving 100% of the identity branch's structural variance."

[Literature] ReNorm



(a) BN **(b)** IN **(c)** ReNorm (ours)

- "BN" model exhibits smaller distances between query and gallery sets of the same identity but is heavily influenced by domain factors, leading to clear domain boundaries.
- Fig.4 (b) shows that IN removes these boundaries but increases the distance between query and gallery sample pairs.

The observations prove that BN can capture more discriminative features but tends to overfit to source domains, while IN can eliminate domain-specific information to enhance generalization but inevitably removes discriminative information.

- Fig.4 (c) shows that the distances between query and gallery sample pairs are much smaller
- Simultaneously, there are no apparent boundaries among target domains, indicating that our method effectively eliminates domain-specific information.

[Literature] CLIP-FDGI

CLIP FDGI: Fine-grained Domain Generative Interchange

- **The Problem:** Purely visual domain gaps (e.g., lighting, extreme camera angles) remain difficult to bridge without computationally heavy pixel-level GANs or feature mixups.
- **The Solution:** Leverages Vision-Language Models (CLIP) to perform "Generative Interchange" in the semantic text space.
- **Mechanism:** Disentangles prompts into [ID Token, *Domain Token*]. By swapping the domain token while keeping the ID token constant, it synthesizes target-domain proxies constrained by an InfoNCE loss.

Comparing with BAU:

- **Strategy:** FDGI augments the training domain distribution virtually using text-driven prompt-swapping. BAU utilizes only the existing multi-source visual domains and explicitly forces distance between them.
- **Philosophy:** FDGI tackles unseen domains by trying to conceptually simulate them during training via the vast prior knowledge of CLIP. BAU tackles unseen domains by regularizing the space to be tightly alignment-focused, hoping the hypersphere uniformity generalizes naturally.

- *Context:* With the rise of foundational models, FDGI maps Re-ID features to contrastive text embeddings to bridge domains visually.
- *Elaboration:* Instead of faking images, FDGI forces the image encoder to disentangle identity from domain
    - by contrasting against synthetically swapped cross-modal prompts.
- *Connection to our work (BAU/Finsler):* FDGI validates our core hypothesis: identity and domain *must* be explicitly disentangled.
    - Just as FDGI uses a learned domain token in language space, our DomainConditionedDriftHead infers a soft domain prior in the visual metric space.
    - We achieve systematic domain disentanglement geometrically, without the extreme computational overhead of multi-modal architectures.

[Model] Omega Regularization Sweep

	mAP		Rank 1		Mean Norm	Max Norm
Euclidean	42.21		42.5			
0.5	41.62	-0.59	42.21	-0.29	0.2636	0.3105
0.75	42.81	+0.6	42.14	-0.36	0.1841	0.2283
1.0	43.07	+0.86	43.57	+1.07	0.1294	0.1589
1.5	43.95	**+1.74**	43.93	+1.43	0.0762	0.1001
2.0	43.43	+1.22	43.5	+1	0.0553	0.0677
2.5	43.69	+1.48	44.14	+1.64	0.0448	0.0531
3.0	43.7	+1.49	44.43	**+1.93**	0.0423	0.0468
5.0	42.23	+0.02	42.93	+0.43	0.0386	0.0439

Small omega norm / mean means either:

1. Asymmetry is not useful in current setup
2. Current optimization setup makes the drift branch too weak to matter

[Model] Attempts at Fixing SLERP

Root Cause: The Arccosine Singularity
Gradient cascades to infinity near identical or diametrically opposed embeddings

$$\frac{d}{dx}\arccos(x) = \frac{-1}{\sqrt{1-x^2}} \xrightarrow{x\to\pm 1} -\infty$$

Solution A: Bounded Linear Extrapolation for arccos
Instead of naïve clamping (results in uninformative zero-gradients at boundaries), switched to tangential linear extrapolation away from singularity.

$$\check{\arccos}(x) = \begin{cases} \arccos(B) - \frac{x-B}{\sqrt{1-B^2}} & \text{if } x > B \\ \arccos(x) & \text{if } |x| \leq B \end{cases}$$

Adopted from SO(3) handling found in PyTorch3D (*acos_linear_extrapolation*)

Solution B: Small-Angle Taylor Expansions
Prevent 0/0 NaNs in the velocity overlap integral via smooth series expansion

$$\frac{\theta}{\sin\theta} \approx 1 + \frac{\theta^2}{6} + \frac{7\theta^4}{360} \quad \text{for } |\theta| < 10^{-3}$$

- When modeling our Finsler drift via SLERP or analytical integration, we noticed immediate model divergence.
    - culprit lies in slerp/analytical finsler distance: we require the angle θ, derived from $\arccos(\cos\theta)$
    - Because our identity features are unit-normalized, positive pairs naturally yield a cosine similarity extremely close to 1.
    - In backpropagation, the derivative of arccps features a $\sqrt{1-x^2}$ in the denominator.
    - As x approaches 1, our gradients explode to infinity. Under mixed-precision, this instantly generates NaNs.

- Previous fixes just clamped the inputs
    - But clamping sets gradients exactly to zero at the threshold, which causes stalled learning.
    - So, we looked to state-of-the-art geometry libraries like PyTorch3D.
    - We implemented a linearly extrapolated arccosine.
        - When the cosine similarity exceeds a strict safety bound B, we freeze the curvature and smoothly follow the tangent line
        - maintaining a stable, non-zero gradient all the way through.

    - B is 1.0 - 1e-4 if fp16, otherwise 1.0 - 1e-5
    - To doubly fortify this, these sensitive angular matrix overlaps are locally forced from FP16 back into strict FP32 precision to grant us the widest representational margin possible.

- Finally, our asymmetry integrals rely heavily on the ratio of θ over $\sin(\theta)$.
    - For extremely small angles—again, highly common during convergence—this creates an unstable $0/0$ division.
    - Instead of a hard, flat mask which ignores local curvature, we adopted operations standardized by Riemannian frameworks like Geoopt.
    - We swap out the direct calculation for its 4th-order Taylor Expansion below a set angular threshold limit.
    - This ensures the geometric traversal stays continuously smooth and computationally bounded, fully eliminating our divergence issues with training collapse during training

[Experiment] Protocol 2 Head to Head

	M+MS+CS → C3		M+CS+C3 → MS		MS+CS+C3 → M		Average	
	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1	mAP	Rank-1
CLIP-FGDI	**44.4**	44.6	**31.1**	**59.4**	**79.4**	**91.3**	**51.6**	**65.1**
ReNorm	43.6	44.7	25.6	55.6	72.7	89.1	47.3	63.1
ReNorm*	42.4	41.5	24.2	53.3	72.6	88.9	46.4	61.2
BAU	42.8	43.9	24.3	50.9	77.1	90.4	48.1	61.7
BAU*	42.21	42.5	23.28	49.70	76.61	89.61	47.37	60.60
Finsler BAU*	43.95	43.93	22.22	48.04	76.47	89.91	47.55	60.63

Sample-level Finsler asymmetry is not strongly supported in DG person ReID; however, domain-structured directional bias remains meaningful and should be modeled as a nuisance factor separated from identity. To continue, we need to consider the following angles:

- *When does asymmetry help*?
- *At what level should it be modeled*?
- *Why does DG training collapse it towards near-Euclidean behavior*?

- Finsler BAU is effectively tied with BAU on the average -> too small to support a strong "asymmetric geometry clearly improves DG person ReID" claim

- Training intentionally suppresses instability in drift branch: reduced drift LR, clipping, omega regularization and removing the drift vector from the alignment loss
- While DG objective itself wants identity invariance -> conflicts with an identity-based directional drift signal

1. When does asymmetry help?

Asymmetric distances (such as Randers/Finsler metrics) are advantageous strictly when the underlying information matching is non-commutative.

- In person Re-ID, metric asymmetry is beneficial when moving from an information-poor domain (e.g., low-resolution aerial probes) to an information-rich domain (e.g., high-resolution ground CCTV).
- It compensates for the fact that feature containment is directional: the features visible in an aerial shot are a subset of those in a close-up, but the reverse is not true.

2. At what level should it be modeled?

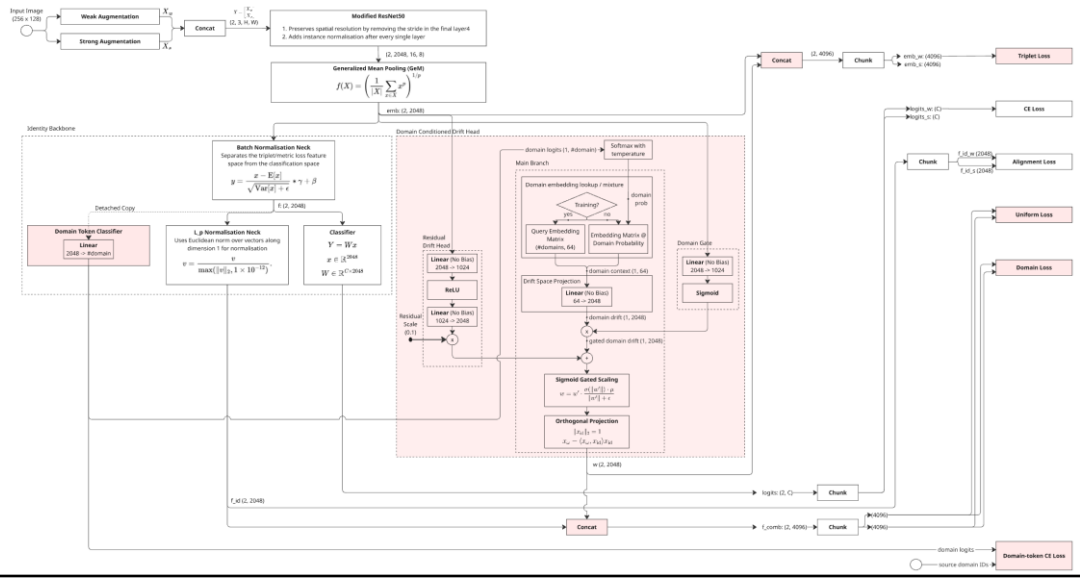
Asymmetry should be modeled at the **domain, view, or modality level**, not the instance level.

- Formulating a Finsler drift vector strictly on a per-identity or per-image basis parameterizes instance-specific noise rather than systematic geometric bias.
- Literature on disentangled representations indicates that nuisance factors (like camera angle, illumination, or sensor modality) are shared across sets of images.
    - *Kulis, B., et al. "Asymmetric metric learning." ICML, 201*
- Thus, the asymmetric drift vector must be decoupled from the identity representation and conditioned on the domain or camera view to act as a proper parallel-transport mechanism across the manifold.

3. Why does DG training collapse it towards near-Euclidean behavior?

- Domain Generalization (DG) objectives—specifically alignment and uniformity constraints (such as those in BAU)—function by explicitly penalizing domain-specific structure to enforce a domain-invariant latent space.
- If an asymmetric drift vector captures domain bias, the DG domain-invariance regularizers will view this drift as a nuisance feature and optimize it toward zero.
- Furthermore, deep neural networks exhibit "simplicity bias" under stochastic gradient descent; if a symmetric Euclidean distance is sufficient to separate identities, the network will discard the more complex Finsler gradient pathways, resulting in a metric collapse to the Euclidean baseline.

[Model] Architecture Proposal: Domain Conditioned Global Drift



- Previously, our *Finsler Drift Head* generated an asymmetric drift vector for every single instance independently.
- Our experiments revealed that Domain Generalization (DG) objectives naturally suppress this.
    - Networks exhibit a "simplicity bias" (*Shah et al., 2020*); if a symmetric metric works well enough, the network zeros out the complex instance-level drift.
- In real deployments (e.g., aerial vs. CCTV), asymmetry is a *domain/view-level property*, not an identity property.
- Therefore, we've redesigned the drift head to explicitly condition the asymmetric geometry on the domain/view structure.

The Foundation: The Identity Pathway

(Point to the top/left of the diagram)

- We maintain our modified ResNet50 (removing the final stride) and GeM pooling (*Radenović et al., 2018*) to preserve fine-grained spatial cues.
- The Batch Normalization neck and L_2 normalization cleanly project the identity into a view-invariant unit hypersphere.
    - **Role:** This branch acts as our rock. It focuses *only* on identity

discrimination.

The Core Innovation: Domain Prediction & Soft Mixtures
Point to the "Domain Token Classifier" (Pink box, left).

- We branched off a *detached* copy of the identity feature into a linear Domain Classifier. This prevents domain-supervision gradients from corrupting the core identity backbone (*Ganin & Lempitsky, 2015*).

Point to "Domain embedding lookup" (Center).

- During training, we know the source domains.
- But at evaluation on an *unseen* target domain, this classifier uses a temperature-scaled softmax to infer a soft probability mixture
    - (e.g., "This looks 70% CCTV, 30% Wearable").
- Instead of generating drift from scratch, we use those probabilities to mix learned domain embeddings.
- This enforces a crucial inductive bias: asymmetric drift is *shared* across identities based on view.

Adaptive Modulation: Gating and Residuals
Point to the "Domain Gate" and "Residual Drift Head".

- Not every image needs full domain drift.
    - A small neural gate dynamically scales the domain context so the model can gracefully fall back on identity for atypical samples (*Hu et al., 2018*).
- We still allow a tiny, per-instance residual correction (scaled down to 0.1) for lighting/pose noise.
    - This keeps the domain structure dominant while providing necessary flexibility.

Strict Disentanglement: Post-Processing
Point to the bottom of the Drift Head (Sigmoid Scaling & Orthogonal Projection).

- The sigmoid-gated scaling controls the total magnitude of the drift vector globally, allowing the model to shrink asymmetry to zero if it doesn't aid a specific batch.
- With the orthogonal projection, we mathematically force the drift vector to be orthogonal specifically to the identity vector
    - It ensures drift is strictly a nuisance/view factor.
    - It absolutely prevents the model from using drift as a shortcut to encode identity (*Kumar et al., 2018*).

Loss Integration & Conclusion
Point to the right side (Loss functions).

- BAU's Alignment loss strictly govern the identity slice to keep it augmentation invariant.
- Domain and Uniformity loss makes the embedding domain invariant.
- Our new Domain-Token CE loss trains the domain predictor.

Asymmetric Triplet: Finally, the full concatenated [Identity + Drift] vector is fed into the Triplet Loss. Here, our asymmetric Finsler distance performs direction-aware hard-negative mining.

Conclusion: By explicitly factoring the embedding into a domain-invariant identity and a domain-conditioned directional drift,
- this architecture solves the instance-collapse problem.
- It allows us to apply structured, generalizable asymmetry to completely unseen camera domains.

Negative samples should come from different domains
- Same identity from different domains using finsler distances
- Inject intuition from the equation
- Start simpler – add constrastive loss terms, then change the model

[Model] Architecture Proposal: Domain Conditioned Global Drift

Why the scaled residual?
Pure domain conditioning assumes every camera angle within a domain is perfectly identical.
- scaled residual (λ=0.1) allows the network to make micro-adjustments for localized nuisances (lighting, specific occlusions)
- while enforcing a strict mathematical bias that the overarching domain structure governs the primary asymmetric geometry
  - prevent the model from collapsing back to instance-level overfitting.

Why does orthogonality prevents shortcut learning?
- Neural networks suffer from "Simplicity Bias" (*Geirhos et al., 2020*)
- With the orthogonal projection, the drift vector is stripped of all identity variance
- Drift branch has to model whatever variance is left to minimize the loss on combined embedding

The Threat of the "Lazy" Network: Neural networks suffer from "Simplicity Bias" (*Geirhos et al., 2020*). They achieve the lowest loss using the easiest features available.
- Because the identity vector (x_{id} (is heavily supervised to perfectly separate individuals, it is the most powerful signal in the network.
- If we just concatenate the drift vector (ω) directly, the easiest way for the network to satisfy the Triplet Loss is to lazily copy the identity features into the drift branch.

We explicitly subtract the projection of ω onto x_{id} :$\omega' = \omega - \langle \omega, x_{id} \rangle x_{id}$.
- The dot product is zero. The drift vector is mathematically stripped of *all* identity variance.
- To minimize the loss on the combined embedding now, the drift branch *has no choice* but to model whatever variance is left—which translates perfectly to camera view, angle, and perspective asymmetry.

SOTA Grounding:

Domain Separation Networks: Pioneered by *Bousmalis et al., 2016, NeurIPS*. They proved that without explicit "Difference Losses"

(orthogonality), private domain branches collapse and simply duplicate shared identity features.

Re-ID Disentanglement: Standard practice to prevent metric corruption when separating nuisance factors (like pose) from core identity (*Karanam et al., 2019, IEEE TPAMI*).

[Literature] Orthogonal Projection Hypersphere

Example 3.6.1 ***Sphere***

On the unit sphere S^{n-1} considered a Riemannian submanifold of $\mathbb{R}^n$, the inner product inherited from the standard inner product on $\mathbb{R}^n$ is given by

$$\langle \xi, \eta \rangle_x := \xi^T \eta. \tag{3.33}$$

The normal space is

$$(T_x S^{n-1})^\perp = \{x\alpha : \alpha \in \mathbb{R}\},$$

and the projections are given by

$$\mathrm{P}_x \xi = (I - xx^T)\xi, \qquad \mathrm{P}_x^\perp \xi = xx^T \xi$$

for $x \in S^{n-1}$.

```
def proju(self, x: torch.Tensor, u: torch.Tensor) -> torch.Tensor:
    u = u - (x * u).sum(dim=-1, keepdim=True) * x
    return self._project_on_subspace(u)
```

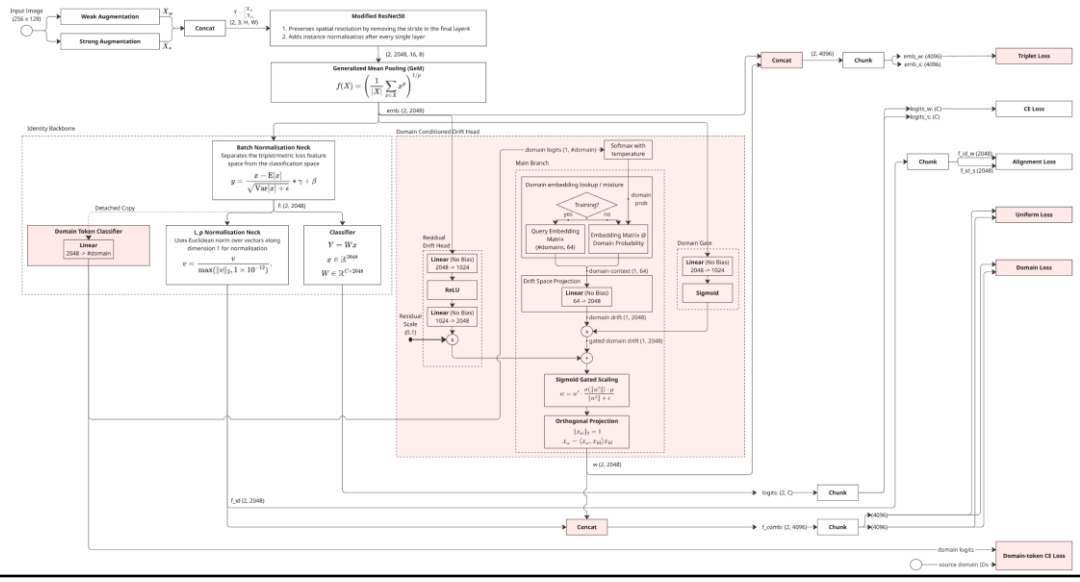
This Week

1. [Administrative] Missing next week's meeting due to person holiday
2. [Model] Revisiting the proposed model from last week
3. [Loss] Changing from RandomIdentitySampler to RandomMultipleGallerySampler
4. [Model] Investigating the Instance Normalization Layers
5. [Loss] Investigating the Uniformity Loss + Identity-based Jaccard Similarity
6. [Model] Plan to migrate Finsler modifications to ReNorm

[Administrative] Next week's meeting cancelled

Reason: Doing a long weekend getaway before starting my Master's thesis. Meeting will resume on 03.04.

[Model] Architecture Proposal: Domain Conditioned Global Drift



Instead of predicting drift purely from the instance features, drift is now composed of: (a) a domain embedding looked up/mixed from source-domain labels, (b) projected and sigmoid-gated by instance features, (c) plus a small residual instance correction (scaled by 0.1)

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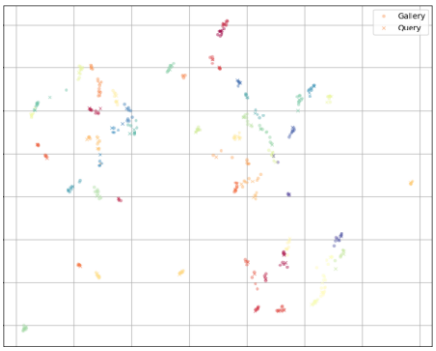
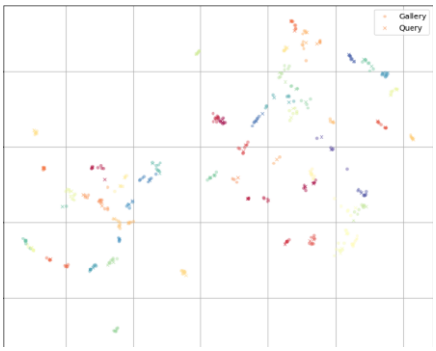
[Model] Architecture Proposal: Domain Conditioned Global Drift

	mAP		Rank 1	
Euclidean	42.21		42.5	
Drift Domain	43.89	+1.68	44.07	+1.57
Drift Instance	43.74	+1.53	44.21	+1.71

Instance Conditioned



Domain Conditioned



RandomIdentitySampler → RandomMultipleGallerySampler

Euclidean Batch-Hard Triplet Loss

$$\mathcal{L}_{BatchHard} = \frac{1}{B}\sum_{i=1}^{B} \max\left(0, \max_{p\in\mathcal{P}(i)} \|f(x_i) - f(x_p)\|_2 - \min_{n\in\mathcal{N}(i)} \|f(x_i) - f(x_n)\|_2 + \alpha\right)$$

Finsler Batch-Hard Triplet Loss

$$\mathcal{L}_{BatchHard} = \frac{1}{B}\sum_{i=1}^{B} \max\left(0, \max_{p\in\mathcal{P}(i)} \{d_f(f(x_i), f(x_p))\} - \min_{n\in\mathcal{N}(i)} \{d_f(f(x_i), f(x_n))\} + \alpha\right)$$

RandomIdentitySampler
Standard PxK sampling over identities and instances – does not use camera IDs, i.e., not guaranteed that the K instances per identity come from K different cameras.

RandomMultipleGallerySampler
Explicitly uses camera IDs: For each identity it picks one anchor, then fills the remaining K-1 with instances from other cameras.

- Looked into the triplet loss formulation -> it already is batch hard
- We had the idea of enforcing the positive to be from a different domain
    - But none of the identities appeared across different domains as the datasets were created separately from each other
    - Which means we can only enforce different camera id's between the x_i and x_p
- The set P(i) is filtered out using the labels and the maximum entry in the distance matrix is taken -> which means the sampler dictates the set
- Proceeded to look into the sampler
    - Found out that

So when an identity has at least K cameras, the K instances are guaranteed to be from K **different** cameras; otherwise it falls back to different instances (possibly same camera).

Rationale
- Intuition is asymmetry is due to camera/view geometry, not per-identity noise.
- To learn that, model must see cross-camera pairs in the batch.
- RandomIdentitySampler can give many same-camera pairs -> little or no

gradient for cross-view asymmetry.

RandomIdentitySampler → RandomMultipleGallerySampler

	mAP		Rank 1	
Euclidean + RandomIdentity (B)	42.21		42.5	
Euclidean + RandomMultiGallery	43.26	+1.05	43.0	+0.5
F-Instance + RandomIdentity	43.74	+1.53	44.21	+1.71
F-Instance + RandomMultiGallery	43.11	+0.9	42.29	-0.21
F-Domain + RandomIdentity	43.89	+1.68	44.07	+1.57
F-Domain+ RandomMultiGallery	44.00	+1.79	44.43	+1.93

Cross-camera sampler improves every reported metric for the domain-conditioned drift.

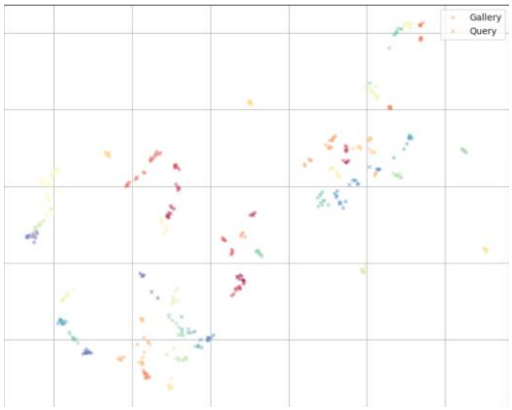
- Intra-identity variation largely due to camera-specific artifacts (lighting, background, viewpoint)
    - Identity-only sampling only gives “easy” same-camera positives
- Triplet and alignment loss get meaningful gradients for view-invariant alignment

Why this hurts the instance-conditioned drift vector

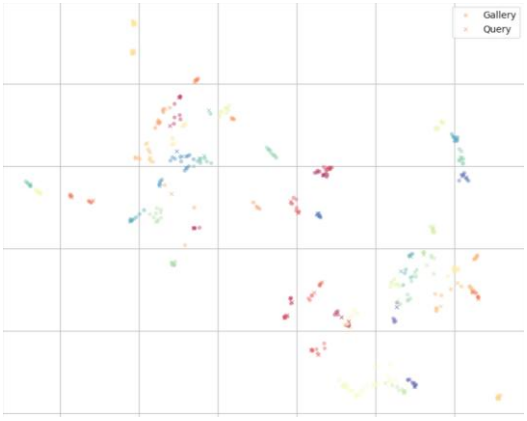
- On RMG, positives are visually far within the same ID.
- The asymmetric distance depends on instance-specific drift for both anchor and positive.
- We’re asking a per-instance channel to reconcile large cross-view appearance gaps together with strong ID alignment / CE / triplet and BAU’s uniform / domain pressures.
- Creates a gradient conflict: triplet + alignment want one consistent notion of “same person” across cameras; F-Instance invites pair- and sample-dependent metric tweaks.

RandomIdentitySampler → RandomMultipleGallerySampler

F-Domain+ RandomMultiGallery



Euclidean Baseline BAU



Do we need the Instance Normalization Layers?

```python
class resnet50(nn.Module):
    def __init__(self, num_classes=0, pretrained=True):
        super(resnet50, self).__init__()
        self.num_classes = num_classes

        # resnet50 backbone
        resnet = torchvision.models.resnet50(pretrained=pretrained)

        resnet.layer4[0].conv2.stride = (1, 1)
        resnet.layer4[0].downsample[0].stride = (1, 1)

        self.base = nn.Sequential(
            resnet.conv1, resnet.bn1, resnet.relu, resnet.maxpool,
            resnet.layer1, nn.InstanceNorm2d(256),
            resnet.layer2, nn.InstanceNorm2d(512),
            resnet.layer3, nn.InstanceNorm2d(1024),
            resnet.layer4, GeneralizedMeanPoolingP(output_size=(1, 1))
        )

        # pooling
        self.pool = GeneralizedMeanPoolingP(output_size=(1, 1))

        # bnneck and classifier
        self.bn_neck = nn.BatchNorm1d(2048)
        init.constant_(self.bn_neck.weight, 1)
        init.constant_(self.bn_neck.bias, 0)
        self.bn_neck.bias.requires_grad_(False)
        if self.num_classes > 0:
            self.classifier = nn.Linear(2048, self.num_classes, bias=False)
```

Two at Once: Enhancing Learning and Generalization Capacities via IBN-Net

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JIA ET AL.: GENERALIZING PERSON RE-ID IN PRACTICE 1

Frustratingly Easy Person Re-Identification: Generalizing Person Re-ID in Practice

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This is a ResNet-IN backbone, which is a result of InstanceNorm being known for style normalization.

- Paper 1 (2018): Introduces IBN-a / IBN-b; ReID is part of the story. IN mixed into ResNet for cross-domain
- Paper 2 (2019): Explicit “IN only in shallow ResNet stages, not in the deepest stage” for domain-generalizable ReI

ReNorm’s core claim is the usual BN–IN tradeoff pushed one step further:

- IN strips domain-specific statistics, which helps domain invariance but can remove identity-relevant signal
- they address that with Remix Normalization + Emulation Normalization and Domain Frozen, not by “more IN.”

IN sits before layer 4, so it shapes the pooled descriptor emb that feeds both the BN neck and drift head

- Preconditions what statistics reach emb
- Removing it is an ablation test of whether **explicit drift replaces implicit static IN**.

Uniformity Loss Requires Symmetric Distances + Identity-Based Jaccard

$$\mathcal{L}_{\text{uniform}} = \log\left(\frac{1}{|\mathcal{I}_{\text{data}}|}\sum_{(i,j)\in\mathcal{I}_{\text{data}}} e^{-2\|\mathbf{f}_i-\mathbf{f}_j\|_2^2}\right) + \log\left(\frac{1}{|\mathcal{I}_{\text{data}}|}\sum_{(i,j)\in\mathcal{I}_{\text{data}}} e^{-2\|\tilde{\mathbf{f}}_i-\tilde{\mathbf{f}}_j\|_2^2}\right), \quad (6)$$

where $\mathcal{I}_{\text{data}} = \{(i,j) \mid i \neq j\}$ is the index set of all distinct pairs within a mini-batch. By

Uniformity assumes a symmetric distance to push embeddings towards uniform coverage of the hypersphere

- Applying it with asymmetric distance might create an ill-defined optimization objective
- Asymmetric points not constrained to the sphere might *invalidate* the usual "uniform on the hypersphere minimizer" story from the original BAU narrative

Uniformity loss should operate on identity-only features with Euclidean distance.

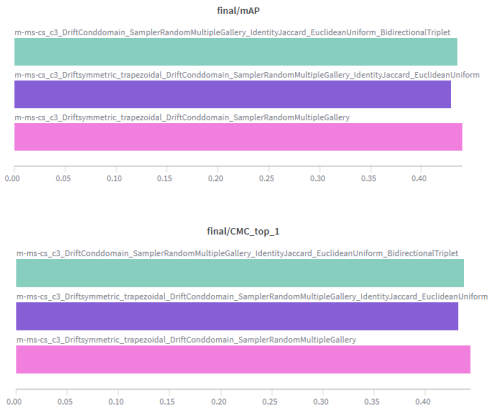
relationship with nearest neighbors [8, 9, 32, 84, 100], we compute the weight as the Jaccard similarity of k-reciprocal nearest neighbors between the augmented sample and the original sample, defined by:

$$w_{ij} = \frac{|\mathcal{R}_k(\tilde{\mathbf{f}}_i) \cap \mathcal{R}_k(\mathbf{f}_j)|}{|\mathcal{R}_k(\tilde{\mathbf{f}}_i) \cup \mathcal{R}_k(\mathbf{f}_j)|} \in [0,1], \quad (4)$$

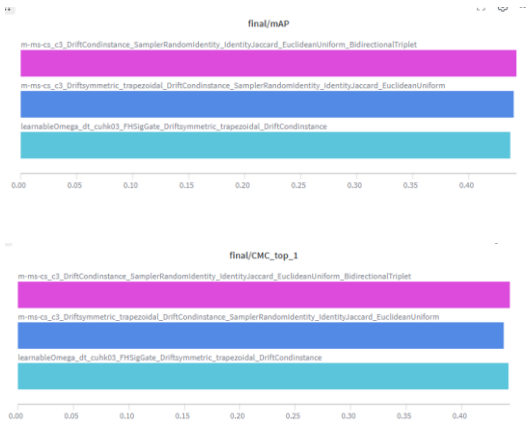
where $\mathcal{R}_k(\mathbf{f}_i)$ is the set of indices for k-reciprocal nearest neighbors within a mini-batch of the feature $\mathbf{f}_i$. Intuitively, a low weight implies that the augmented feature $\tilde{\mathbf{f}}_i$ and the original feature $\mathbf{f}_j$

Uniformity Loss Requires Symmetric Distances + Identity-Based Jaccard

Domain Conditioned



Instance Conditioned



General Critiques on Current Architecture Proposal

1. Adding a 2048-dimensional drift vector doubles the embedding dimensionality
   - → *Any marginal improvement could be an artifact of increased capacity rather than asymmetric geometry*
   - → To be monitored in ablation
2. Domain embedding only has 64 dimensions, projected to 2048
   - → *Rank-64 bottleneck, i.e., domain-conditioned drift can only span a 64 dimensional subspace. Sufficient to capture meaningful asymmetry between domain shifts?*
   - → With $S = 3$ source domains, the domain embedding matrix $E \in \mathbb{R}^{3\times \mathrm{d}}$ is projected through $W \in \mathbb{R}^{\mathrm{d}\times 2048}$ to produce the domain drift basis $M = EW \in \mathbb{R}^{3\times 2048}$. The rank of M is bounded by $\min(3, d)$. For any $d \geq 3$, the domain-prior subspace is at most rank 3.
3. The drift vector is conditioned on the domain-id of the identity, while no cross-domain identities exist
   - → Sampler can only ensure that positives come from different cameras; metric learning signal from triplet loss does not complement modified architecture
   - → Modify the conditioning to be based on camera-id and not domain-id

The sigmoid gate $\sigma(\text{Linear}(x)) \in \mathbb{R}^{2048}$ applies element-wise modulation per instance, but this does not increase the rank of the domain prior itself — it only rescales the fixed basis vectors per-sample. Sweeping $d \in \{32, 64, 128\}$ therefore changes the number of dead parameters and the conditioning of the projection matrix, but not the representational capacity of the domain prior.

Next Week

1. [Model] Finslerized ReNorm
2. [Sweep] (Ongoing) Softmax temperature and domain token loss weight
3. [Sweep] Omega regularizer scalar weight
4. [Sweep] Scalar multiplier for residual correction term
5. [Ablation]
6. [Ablation]

Questions

1. [Administrative] Signing up for Master Thesis